Python Day 1–3 Practice Problems

Loop-focused practice • Questions only • Difficulty: Low, Medium, and High

Instructions: Every problem should use at least one `for` loop or `while` loop. Use the Python concepts covered through Day 3, including variables, input, strings, lists, operators, conditions, `break`, `continue`, and `match`/`case` where requested.

Low Difficulty

1. Print numbers 1–10
Use a `for` loop and `range()` to print the numbers from 1 through 10.

2. Print even numbers
Print all even numbers from 2 through 20.

3. Countdown
Use a `while` loop to print a countdown from 10 to 1. Print “Blast off!” after the loop.

4. Repeat a greeting
Ask the user for their name. Use a loop to print “Hello, <name>!” five times.

5. Sum from 1 to 5
Use a loop to add the numbers from 1 to 5. Print the final total.

6. Print characters in a word
Ask the user for a word. Use a `for` loop to print one character per line.

7. Print items in a list
Create a list containing five fruits. Use a loop to print each fruit.

8. Multiplication table
Ask the user for a number. Print its multiplication table from 1 to 10. For example, if the user enters 4, display lines such as “4 x 1 = 4” through “4 x 10 = 40”.

Medium Difficulty

9. Count vowels
Ask the user for a word or sentence. Use a loop to count how many vowels it contains: a, e, i, o, and u.

10. Add numbers until zero
Repeatedly ask the user for an integer. Add each number to a total. Stop when the user enters 0, then print the total.

11. Positive-number counter
Ask the user to enter five numbers. Use a loop to count how many of them are positive.

12. Find the largest number
Ask the user to enter five integers. Use a loop to find and print the largest number entered.

13. Skip negative values
Use this list: `[4, -2, 8, -5, 0, 12, -1]`. Use a loop and `continue` to skip negative numbers. Print only zero and positive values.

14. Stop at a secret word
Keep asking the user to enter a word. Stop the loop with `break` when they enter “stop”. Do not print “stop” as part of the normal output.

15. Password attempts
Give the user three attempts to enter the password “python123”. If correct, print “Access granted” and stop the loop. If all attempts are used, print “Access denied”.

16. Simple menu with match
Repeatedly display this menu until the user chooses 4: 1. Say Hello; 2. Print Numbers 1–5; 3. Show Python Message; 4. Exit. Use `match`/`case` to respond to the selected option.

High Difficulty

17. Average of entered scores
Ask the user how many test scores they want to enter. Use a loop to collect that many scores, calculate the average, and print “Pass” if the average is 50 or higher; otherwise print “Fail”.

18. Number guessing game
Set `secret_number = 7`. Use a `while` loop to let the user keep guessing until they get the correct number. For each incorrect guess, print “Too high” or “Too low”. When correct, print how many attempts were needed.

19. Shopping-list total
Ask the user to enter five item prices. Use a loop to calculate the total cost. Apply a 10% discount if the total is greater than 100, then print the final total formatted to two decimal places.

20. Student grade report
Create this dictionary: `{“Ava”: 82, “Liam”: 45, “Noah”: 91, “Mia”: 68, “Zoe”: 50}`. Use a loop to print each student’s name, score, and result: A for 90 or above, B for 70–89, C for 50–69, and Fail below 50.